

Using relative path:

Input/input1.txt

Output/output1.txt

Handle client crash or idle client:

Will maintain last seen for every operation if current time – last seen > 10 min client will revoke all resources and close the file

Create:

- File already exists.
- Concurrent create
- Directory doesn't exist
- Permission issue – only output file can be created
- Crash during creation – metadata written but not initialised ---can run a manual routine check up on disk which can delete such zombie files
- Duplicate request due to network error - Use unique request IDs and maintain a server-side cache of processed requests.

If the same request ID appears again, return the stored response instead of re-executing the operation. ----- If we store all requests memory will full this we will maintain a cache which store request for limited time or in limited number

Delete

- Deleting a file that doesn't exist
 - File currently open by some other client
 - Concurrent write + delete – delete request while write in progress
 - Duplicate rpc request
 - No directory delete request allow
- If any other client is using the file, error msg should return not allow to close the file

Open + Read

- File doesn't exist
- One client should open file once
- Input file should be read only and output files can be read and write both
- 2 client tries to open at same time
- Server restart ---- if fd becomes invalid
- Client crashes server must release the lock
- Deletion of file while it is open is not allowed---- it is been handled in delete function
- File close on failure --- If something fails after open, file should be closed
- One client is writing file, open request should wait in queue till it over
- Update lastseen on every operation

Reader and writer should wait in queue if output file is being written by someone else.

Close + Write AFS style.

- Write or updates are only allowed in output file.
- Input files should only close.
- Write or close should wait if a writer is writing. Waiting should be in queue.
- Handle duplicate RPC request.
- Update only if dirty bit is changes by client – client has changed the file.
- Server crash during write – changes should be revoke
- Client crash without closing
- Writes should be append
- Writing in different order of arrival is allowed.
- Atomicity --- Handle the case when server crash while writing in finale output file --- Use .tmp file